Case Study 8: Multi-Stage Examination Eligibility System

1. Problem Statement

Develop a Python program to determine whether a student is eligible to attend the final examination based on attendance, laboratory completion, and internal assessment marks.

2. Objectives

- To accept student details from the user.
- To use conditional statements.
- To check examination eligibility.
- To display the eligibility result.

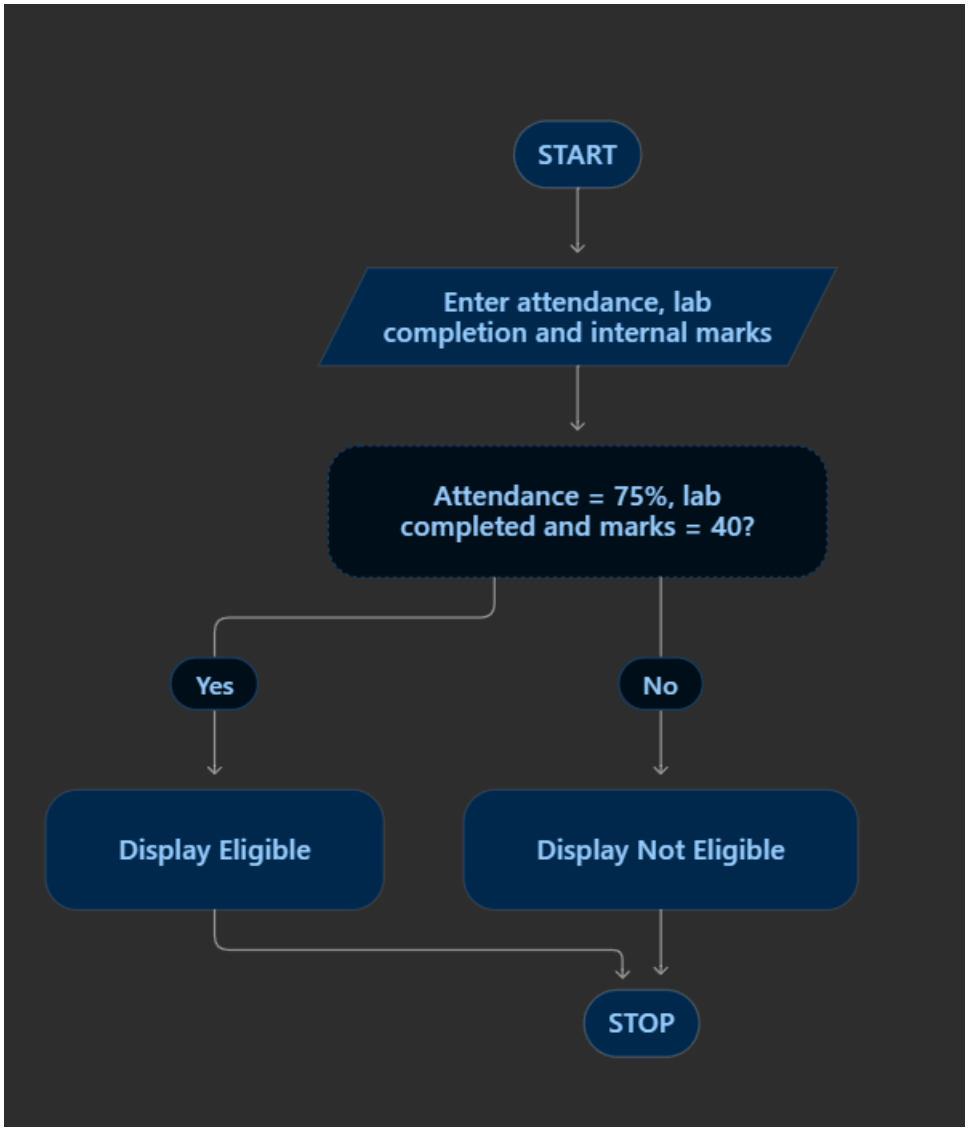
3. Assumptions

- Minimum attendance required is 75%.
- All laboratory work must be completed.
- Minimum internal assessment marks required are 40 out of 100.
- These are assumed values for this program; actual college rules may differ.

4. Algorithm

1. Start.
2. Enter attendance percentage.
3. Enter whether laboratory work is completed.
4. Enter internal assessment marks.
5. Check whether all three eligibility requirements are satisfied.
6. If all conditions are satisfied, display “Eligible”.
7. Otherwise, display “Not Eligible”.
8. Stop.

5. Flowchart



6.Pseudocode

START

INPUT attendance

INPUT lab_completion

INPUT internal_marks

IF attendance >= 75 AND

lab_completion == "yes" AND

internal_marks >= 40 THEN

PRINT "Eligible for Final Examination"

ELSE

PRINT "Not Eligible for Final Examination"

END IF

STOP

7.Source code

Examination eligibility system.py

```
attendance = float(input("Enter attendance percentage: "))
lab = input("Is laboratory work completed? (yes/no): ")
marks = float(input("Enter internal assessment marks: "))

if attendance < 0 or attendance > 100 or marks < 0 or marks > 100:
    print("Invalid input!")

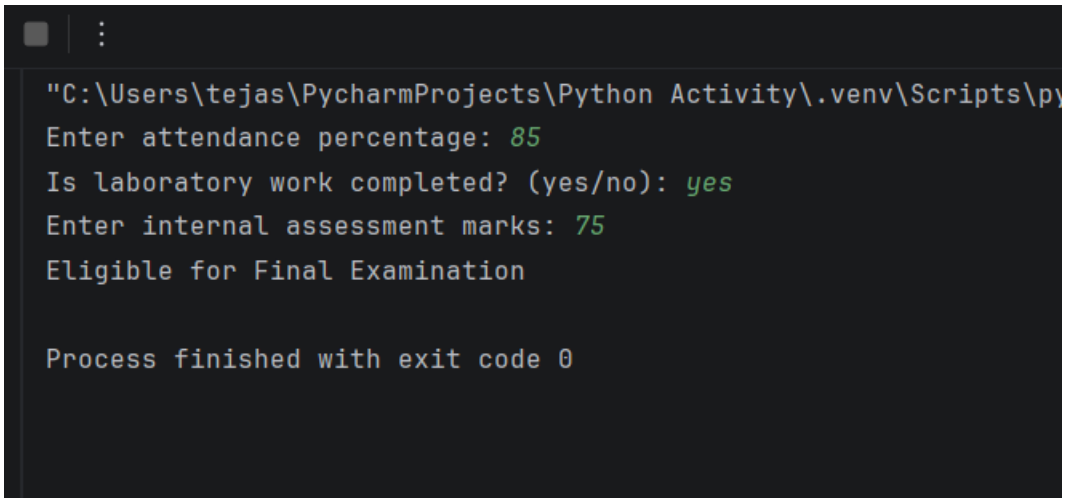
elif attendance >= 75 and lab == "yes" and marks >= 40:
    print("Eligible for Final Examination")

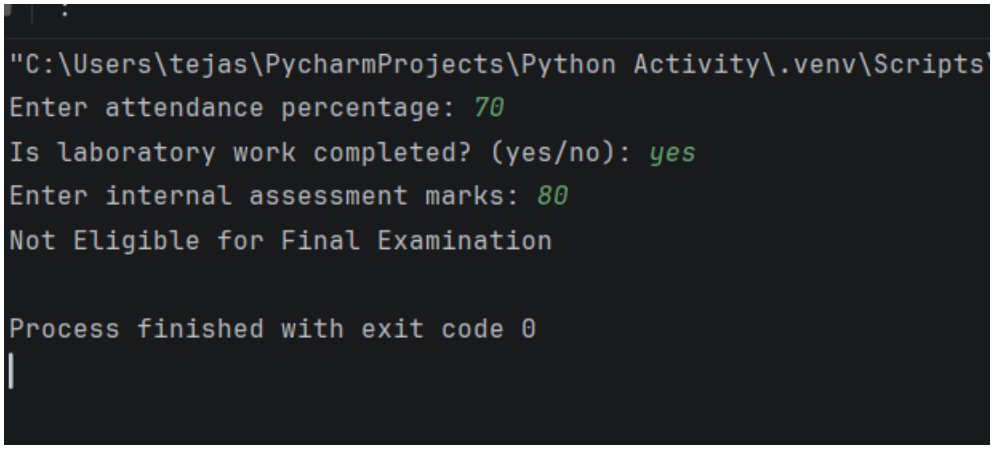
else:
    print("Not Eligible for Final Examination")
```

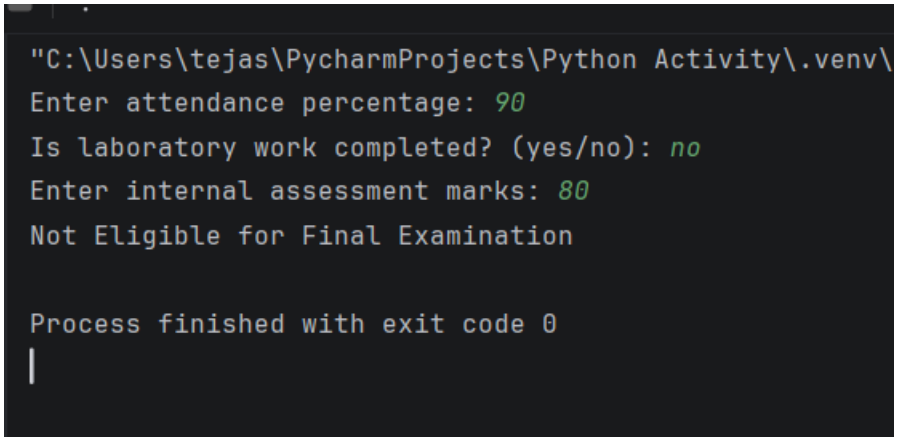
8.Test cases

Attendance	Lab completed	Internal marks	Expected result
85%	Yes	75	Eligible
70%	Yes	80	Not eligible
90%	No	80	Not eligible

9.Output







10. Conclusion

The Multi-Stage Examination Eligibility System uses user input and conditional statements to determine whether a student meets the requirements for attending the final examination.